

Marginal-price support of electric-bus fleets with full battery replenishment

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Abstract

Electric-bus charging combines continuous energy decisions with indivisible service assignments. We ask whether a feasible fleet schedule is a best response to the marginal electricity prices generated by its own load. An exact, fully replenished two-service construction has a positive gap between physical and convexified planning, which persists with charging losses, a reserve and a finite connector. Replication can make the absolute planning gap vanish while whole-operator regret remains positive along a subsequence; regret per reserved pair vanishes. For two depot variants of a public timetable, a fleet-cardinality and energy argument gives analytical lower bounds stronger than the reported iterative bounds. Joint reporting of physical cost, hull cost and own-price regret shows what existing computations establish and what remains unresolved. The public intervals do not establish a positive planning gap, and regret of a returned incumbent does not establish regret at an optimal fleet. These distinctions guide further computational evaluation.

Keywords: electric-bus scheduling; convex-hull pricing; marginal tariffs; charging coordination; nonconvex scheduling.

1 Introduction

An electric-bus operator must cover mandatory passenger services while assigning vehicles and scheduling charging. A tariff can shift charging toward cheaper periods, but a fleet large enough to affect aggregate electricity demand also changes the marginal cost behind that tariff. Even with continuously adjustable charging, its service assignments remain indivisible. The resulting question is whether a physically realizable fleet schedule minimizes the operator’s bill when evaluated at the marginal prices associated with that schedule’s load.

We study this question using complete fleet plans. Each plan includes all services, vehicle movements, charging and operating cost. Convexifying their joint cost and load vectors provides a useful planning lower bound, but a point in that hull is generally not an executable daily dispatch. Charging a convex supply cost at the mean load also differs from taking the mean cost of randomized physical dispatches. Maintaining these distinctions is essential both for interpreting a pricing failure and for assessing computational shortcuts.

The contribution has two parts. First, a small continuous-charging construction makes the obstruction explicit under full terminal replenishment. A perturbation permits a positive reserve, charging losses and one finite connector. A replication calculation then separates an aggregate planning gap from the incentives of a whole-fleet operator and of participants with individually reserved resources. These are application-specific exact results within established nonconvex pricing

theory. Second, an energy argument strengthens the public timetable’s lower bounds, and existing computations are organized around physical planning cost, hull cost and own-price regret. This makes the computational evidence answer the same question as the analytical model.

The computational scope is two small synthetic diagnostics and two depot variants of one public timetable. The public bounds remain open; they do not establish a positive nonconvexity gap or the absence of a supporting physical plan on that timetable. The main paper reports those limits alongside the exact mechanism. A separate development supplement preserves exploratory reuse and solver-start comparisons, whose budgets, incomplete intervention records and software failures prevent general acceleration claims. Broader computational validation remains part of the research program.

2 Related work

Electric-bus scheduling jointly assigns service to vehicles and tracks battery and charging decisions. Wu et al. study multi-depot scheduling with grid objectives; de Vos et al. model partial charging and station capacity; Parmentier et al. scale route and recharge scheduling by column generation [Wu et al., 2022, de Vos et al., 2024, Parmentier et al., 2023]. These works establish operational roles for duty construction, energy limits, and charging resources. Here we ask how a complete fleet plan relates to the convex hull of complete plans under a common convex supply-cost model. A duty-column relaxation need not equal this whole-fleet hull, and a mean load in the hull need not be executable as one day’s bus schedule.

Public timetable and movement data make this distinction testable without fixing the economics. Sistig et al. generate bus and crew schedules for twenty German transport networks from public GTFS inputs; their versioned dataset includes routes, trips, stops, possible deadheads, and schedules [Sistig et al., 2025a,b]. We use one Hildenbrand timetable in two restricted depot variants. Source distances, times, and bus parameters are combined with EGG’s declared energy, battery, reserve, charger, horizon, and synthetic convex supply-cost assumptions; the dataset supplies no EGG market-supply function. These remain small development studies on one timetable and two depots, with public global bounds open. They establish neither an equal-quality speedup nor a learned-proposal benefit.

Nonconvex electricity-market research explains why marginal prices can fail to support dispatch with indivisible operating decisions. Gribik et al. compare unit-commitment pricing rules and discuss energy uplift when no energy-price vector supports selected quantities [Gribik et al., 2007]; O’Neill et al. construct prices from a mixed-integer model and associated linear program [O’Neill et al., 2005]. Convex-hull pricing studies uniform prices through convexification: Schiro et al. discuss its formulation and implementation, and Hua and Baldick give a convex primal formulation [Schiro et al., 2016, Hua and Baldick, 2017]. Madani et al. compare convex-hull, integer-programming, and European-style pricing with nonconvex demand bids; Andrianesis et al. compute convex-hull prices by Dantzig–Wolfe decomposition and column generation [Madani et al., 2018, Andrianesis et al., 2022]. Hümbes et al. characterize linear-price support for decentralized problems with integralities and complete linear feasible-set descriptions [Hümbes et al., 2022]. These are precedents for convex-hull pricing and price-support questions; we use the same general supporting-price logic for a stated fleet set and supply-cost function. Our own-price best-response test is conditional on that model, not a general market-clearing theorem or settlement design; fleet regret is not an uplift payment under these market rules.

Aggregation results also require their own scale and participant definitions. Starr’s classical quasi-equilibrium result gives an aggregation perspective for economies with nonconvex preferences [Starr, 1969]. Our replication construction is proved directly for its specified fleet schedules: the aggregate planning gap tends to zero while whole-operator regret can remain positive, whereas per-bus and individually reserved-pair incentives vanish. A Shapley–Folkman aggregation argument does not by itself prove this exact replication result or force the absolute regret of one composite operator to vanish. The distinction is between aggregating many separately bounded participants and enlarging a single participant whose opportunity set remains indivisible.

Price coordination for flexible EV charging is established. Gan et al. model each vehicle’s charging profile as a continuous, rate-bounded vector with fixed energy over an availability window and give a price-update protocol converging for a convex aggregate-load objective [Gan et al., 2013]. This coordinates charging once availability and energy needs are fixed. Bus schedules additionally assign trips to vehicles, changing charging windows and the joint feasible load set. Convex charging coordination therefore does not imply own-price support for a discrete complete bus schedule.

Fixed-schedule fleet counting has a classical analogue in Dantzig and Fulkerson’s minimum-tanker problem [Dantzig and Fulkerson, 1954]. Such flow reductions count vehicles covering fixed movements, but do not certify battery states, shared-charger feasibility, or price support. We separate minimum-fleet flow diagnostics, charging replay, and convex-hull economic comparisons.

3 Complete fleets and marginal-price support

The unit of optimization is a *complete* fleet plan, rather than an individual vehicle duty. Fix a timetable, allowed directed movements, a charging horizon, and the equipment and energy assumptions. Let $\mathcal{X} \neq \emptyset$ contain precisely the plans that cover every mandatory service once, assign the services to used buses, route those buses through allowed movements, respect battery and charger constraints, and restore every used bus to its required terminal inventory. For $x \in \mathcal{X}$, write $c(x)$ for intrinsic operating cost and $e(x) \in \mathbb{R}_+^T$ for grid energy bought in the T market intervals. An unused bus contributes neither charging nor cost. We use the same $\mathcal{X}$, c , and e for physical planning, price response, and convexification. Finitely many discrete duties with bounded continuous charging give a compact $\mathcal{X}$ in the cases below; assume more generally that its image $\{(e(x), c(x)) : x \in \mathcal{X}\}$ is compact.

In the timed physical model, each used path starts at battery inventory B . A service or directed movement consumes its declared energy in chronological order. Inventory remains between reserve r and capacity B at every event. A depot charging session buying q grid kWh adds ηq inventory, where $0 < \eta \leq 1$, within its availability window and bus-power limit. Concurrent sessions obey the shared power or connector limit. Each path finishes at inventory B by the declared horizon. In the homogeneous single-connector formulation, charging intervals are split at all visit and market boundaries. If interval k has duration d_k and power P_k , its charges obey $0 \leq z_{mk} \leq P_k d_k y_m$ and $\sum_m z_{mk} \leq P_k d_k$, where y_m indicates the selected visit. Serial allocation of z_{mk}/P_k hours to the visits realizes these totals when switching time is zero, efficiency is constant, and charging does not taper. The worked constructions below assume zero travel cost and no unlisted travel or auxiliary load; the public case declares its movement energy separately.

Let $F : \mathbb{R}_+^T \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper, closed, convex supply cost that admits a finite differentiable convex extension to a neighborhood of the convex hull of feasible loads. The physical and complete-fleet-hull values are

$$\begin{aligned} D &= \min_{x \in \mathcal{X}} \{c(x) + F(e(x))\}, \\ CH &= \min_{(e,c) \in \text{conv}\{(e(x), c(x)) : x \in \mathcal{X}\}} \{c + F(e)\}, \quad \Delta = D - CH \geq 0. \end{aligned} \quad (1)$$

The hull mixes whole feasible plans, including their charger constraints; its mean load is a pricing and bounding object, not necessarily a dispatch. For a posted price p , define the complete-fleet response $V(p) = \min_{x \in \mathcal{X}} \{c(x) + p^\top e(x)\}$. At a physical plan x , $p_x = \nabla F(e(x))$ and

$$r(x) = c(x) + p_x^\top e(x) - V(p_x). \quad (2)$$

The alternative in $V(p_x)$ takes the posted price as fixed. This is the appropriate support test for a price-taking fleet, even though its original load determined p_x . A strategic fleet instead compares bills such as $c(y) + \nabla F(e(y))^\top e(y)$ across its alternatives y and anticipates the resulting price change. That objective differs from both $c(y) + p_x^\top e(y)$ and the planner's $c(y) + F(e(y))$; it needs a specified settlement rule and strategic game. The present price-support and compensation statements make no claim about strategic behavior [O'Neill et al., 2005, Gribik et al., 2007, Gan et al., 2013].

Proposition 1 (Complete-fleet support). *Under the common-model and attainment assumptions above, every physical $x \in \mathcal{X}$ satisfies*

$$r(x) \geq c(x) + F(e(x)) - CH \geq D - CH = \Delta. \quad (3)$$

Moreover, $D = CH$ if and only if some physical plan is a best response at its own marginal price. If $D = CH$, every physical planner optimizer has this property.

The proof in Appendix A is a specialization of classical nonconvex price-support and convexification arguments [Starr, 1969, O'Neill et al., 2005, Gribik et al., 2007, Hua and Baldick, 2017, Schiro et al., 2016]. It says nothing about whether an iterative price adjustment converges or which optimum an algorithm selects. Attainment matters in the converse: equality of infima alone does not produce a physical supporting plan.

The accounting identity also requires a stated deviation right. In this paper one fleet controls all buses and its alternative must be a member of the same $\mathcal{X}$. Separate operators using shared chargers would need reserved capacities, trades, or congestion rights before a unilateral opportunity set could be defined. For the single fleet, put $F^*(p) = \sup_{u \geq 0} \{p^\top u - F(u)\}$ and define

$$\text{LOC}_{\text{fleet}}(x; p) = c(x) + p^\top e(x) - V(p), \quad (4)$$

$$\text{LOC}_{\text{supply}}(x; p) = F(e(x)) - p^\top e(x) + F^*(p). \quad (5)$$

Whenever $F^*(p)$ is finite, both terms are nonnegative and their sum is $c(x) + F(e(x)) - [V(p) - F^*(p)]$. At a hull-optimal price and a physical planner optimizer, strong Fenchel duality makes that sum Δ ; the fleet and supply portions depend on the chosen common price. At p_x , supply LOC is zero and fleet LOC equals $r(x)$. No budget-balance or payment institution follows from this arithmetic.

For quadratic supply $F(e) = a^\top e + \frac{1}{2}e^\top Qe$ with $Q \succeq 0$, write $\|z\|_Q = (z^\top Qz)^{1/2}$. There is a useful complement to the lower bound. Let $R_Q = \max_{x,y \in \mathcal{X}} \|e(x) - e(y)\|_Q$ and $H(x) = c(x) + F(e(x)) - CH$. Then

$$H(x) \leq r(x) \leq H(x) + R_Q \sqrt{2H(x)}. \quad (6)$$

Thus a small cost gap controls own-price regret only together with the curvature-weighted range of feasible fleet loads. The proof permits singular Q and boundary loads and appears in Appendix A.

For numerical certification, valid bounds $D \in [L_D, U_D]$ and $CH \in [L_{CH}, U_{CH}]$ imply $\Delta \in [L_D - U_{CH}, U_D - L_{CH}]$. A replay-feasible physical plan supplies an upper bound on D ; a mixture of replay-feasible complete plans supplies an upper bound on CH . At any price with finite $F^*(p)$, a global price-response lower bound $\underline{V}(p) \leq V(p)$ gives the hull lower bound $\underline{V}(p) - F^*(p)$. A nearly solved restricted column pool alone cannot supply that global lower bound. In the separable implementation, $F(e) = \sum_t (a_t e_t + b_t e_t^2/2)$ and $b_t \geq 0$; the conjugate term is $[p_t - a_t]_+^2/(2b_t)$ when $b_t > 0$, and is finite at $b_t = 0$ only for $p_t \leq a_t$. These are standard convexification and Fenchel certificates over complete plans, rather than a new general convex-hull pricing method [Hua and Baldick, 2017]. Section 6 compares two specified market states.

4 Exact replenished-fleet constructions

Consider two mandatory depot-to-depot services. Service A occupies hour $[0, 1]$ and service B occupies $[2, 3]$; each consumes 15 kWh. At most two identical buses may be used. Each used bus starts at its full 20 kWh capacity and must return to 20 kWh by hour 4. The early charging window is $[1, 2]$, with 10 kW per bus and in total. The terminal window is $[3, 4]$, with 30 kW per bus and in total and enough connectors. Charging is continuous and lossless; reserve, travel cost, deadhead energy, auxiliary load, switching time, and taper are zero. No charging is available outside these windows. The terminal charge is physical replenishment, so its energy and supply cost are included.

There are only two unlabeled service partitions. If one bus covers A and B, it reaches 5 kWh after A and needs at least 10 kWh before B. The early power limit fixes its early grid energy at $x = 10$ kWh; it buys 20 kWh terminally. If two buses cover one service each, the A bus buys any $x \in [0, 10]$ kWh early and $15 - x$ kWh terminally, while the B bus buys 15 kWh terminally. Every used bus returns full. Both structures therefore buy exactly 30 kWh, and initial inventories cannot subsidize the horizon.

Let a used bus cost $f = 7$ synthetic units and let grid supply cost be

$$F(x, z) = 4x + \frac{x^2 + z^2}{10}, \quad x, z \geq 0. \quad (7)$$

The one-bus physical cost is $7 + F(10, 20) = 97$. The two-bus branch has cost $14 + F(x, 30 - x)$, minimized at $x = 5$ with value 99. Yet the complete-fleet hull has a lower objective.

Proposition 2 (Replenishment does not ensure price support). *For this physical model,*

$$D = 97, \quad CH = \frac{7591}{80} = 94.8875, \quad \Delta = \frac{169}{80} = 2.1125. \quad (8)$$

The physical planner optimum has own-price regret 13. At a common hull-optimal price its fleet lost opportunity is zero and its supplier lost opportunity is $169/80$.

The hull optimizer mixes the one-bus plan $(x, z) = (10, 20)$ with weight $27/40$ and the two-bus plan $(0, 30)$ with weight $13/40$. Its mean early load is $27/4$ kWh. Both components are physical; their mean load and mean operating cost are a convexification, not a fractional bus schedule. Running the two plans on different days gives mean system cost 99.275, because averaging physical supply costs differs from evaluating the convex supply cost at their mean load. The exact hull and price calculations are in Appendix A.

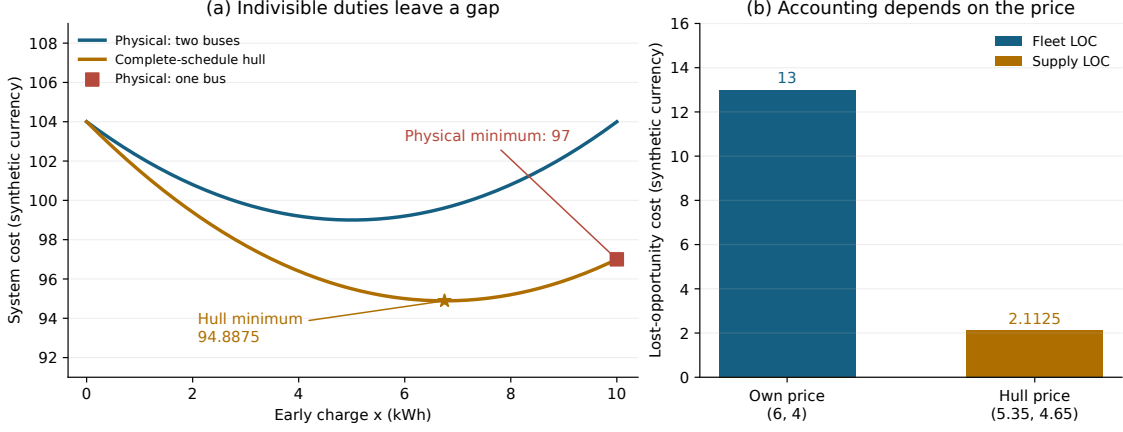


Figure 1: Exact physical branches, complete-fleet hull, and lost-opportunity accounts in the synthetic two-service case. Prices are synthetic units/kWh; the hull mean is not an executable daily fleet.

At the physical optimum, the own marginal price is $(6, 4)$ synthetic units/kWh. The one-bus private objective is 147, while the two-bus plan charging entirely terminally has private objective 134. The resulting 13-unit regret is a fixed-price incentive to change the whole fleet. The hull price is $(5.35, 4.65)$: the two plans supporting its mean tie at private objective 153.5. At that price, the physical one-bus optimum has no fleet lost opportunity, but the supplier term is $169/80$. These two diagnostics use different prices and answer different questions.

The binding 10 kW early limit is a fragile nominal parameter: at the same 20 kWh capacity, a positive reserve or charging loss can remove the one-bus branch. A nearby construction allows those constraints while retaining a gap. Keep two 15 kWh services, battery capacity 20, and (7), but use reserve $r = 1$ kWh, grid-to-battery efficiency $\eta = 19/20$, 12 kW individual and shared early power, and a single 30 kW terminal connector. The one-bus plan buys $220/19$ grid kWh early and 20 terminally, with battery states $20 \rightarrow 5 \rightarrow 16 \rightarrow 1 \rightarrow 20$. Its early power has $8/19$ kWh of headroom. The supporting two-bus plan buys $30/19$ early and 30 terminally; the two terminal sessions take $9/19$ and $10/19$ hour and fit sequentially on the single connector. Every used bus again finishes full.

Proposition 3 (A positive-gap neighborhood). *Under the modified parameters,*

$$D = \frac{38527}{361}, \quad CH = \frac{2987911}{28880}, \quad \Delta = \frac{94249}{28880} > 0. \quad (9)$$

With the 30 kW terminal connector fixed, strict feasible-headroom and optimality inequalities preserve a positive gap in an open neighborhood of the stated reserve, efficiency, and early power.

The interval and strict-inequality proof is in Appendix A. The neighborhood is conditional on this hardware model's zero switching time, constant efficiency, and lack of taper. For perspective on connector adequacy, if the original lossless case has one terminal connector of power $K \leq 30$ kW, it is infeasible for $K < 20$, has zero gap for $20 \leq K \leq 23.5$, and has a positive gap for $23.5 < K \leq 30$. Thus the presence of one connector alone says nothing about price support.

4.1 Replication and the participant scale

Replicate A and B to n simultaneous copies each. Give the fleet n early 10 kW connectors and n terminal 30 kW connectors. Battery and service energy per bus stay fixed; supply cost scales with demand as

$$F_n(E, L) = 4E + \frac{E^2 + L^2}{10n}. \quad (10)$$

If m buses each cover an A–B pair, $n - m$ A-only and $n - m$ B-only buses complete the services. There are $2n - m$ used buses, and physical early energy ranges over $[10m, 10n]$. Terminal sessions have an explicit connector-feasible assignment: paired buses take m connectors, while each A-only/B-only pair can share one of the other $n - m$ terminal connectors sequentially. All buses return full.

Proposition 4 (A vanishing gap with aggregate regret). *The complete projected hull has lower operating-cost boundary $14n - 0.7E$ and optimum $CH_n = 7591n/80$. Every physical optimum takes an integer m nearest to $27n/40$; both neighboring integers are retained at a tie. With $\delta = m - 27n/40$,*

$$\Delta_n = \frac{20\delta^2}{n} \leq \frac{5}{n}. \quad (11)$$

Along $n = 40k + 1$, the whole-operator own-price regret is

$$r_n = \frac{351}{40} + \frac{169}{40n} \longrightarrow \frac{351}{40}, \quad \Delta_n = \frac{169}{80n} \longrightarrow 0. \quad (12)$$

For multiples of 40, both quantities are exactly zero.

The operator in (12) can rearrange the whole growing fleet while taking its posted price as fixed. Its regret per used bus, and regret divided by physical cost, vanish. A different institution gives each of n operators one A/B pair plus a reserved early and terminal connector. Each then chooses either the one-bus plan or a two-bus plan with $x \in [0, 10]$; these separate deviations are physically compatible under the reserved rights. At the original fleet optimum's marginal price, each pair's regret is at most $20/n$, and their sum equals the whole-fleet regret in this symmetric construction. Along $n = 40k + 1$ a dissatisfied one-bus operator has regret $13/n$, although the aggregate tends to $351/40$. The scaling example consequently distinguishes absolute cost gap, whole-operator incentive, and per-participant incentive without asserting a result for arbitrary shared-charger games. At $n = 20$, the two tied planner optima have distinct own-price regrets, 7 and 14.

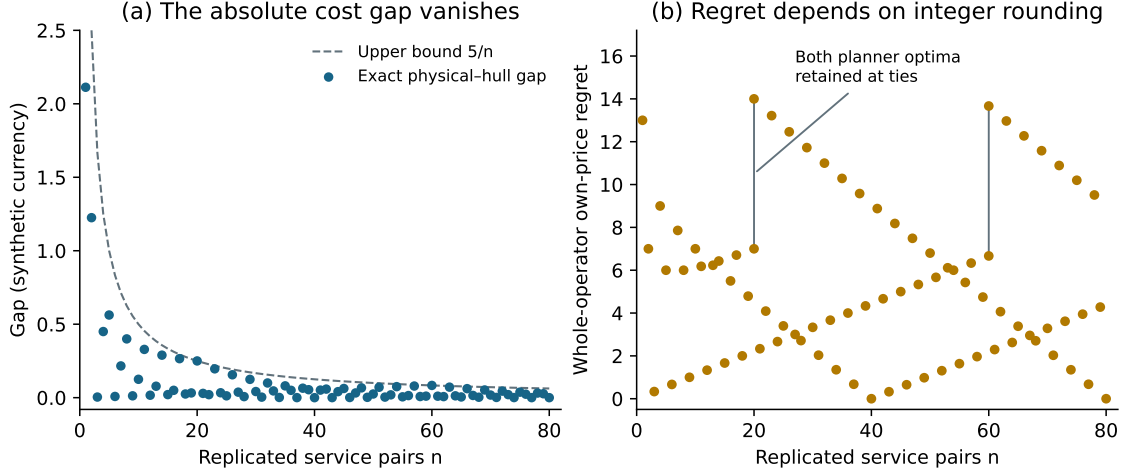


Figure 2: Exact replication for $n = 1, \dots, 80$, including both physical optima at ties. The planning gap follows the $5/n$ bound while whole-operator own-price regret varies with integer rounding. Values are synthetic units, not observations from a bus system.

5 A public timetable: feasible fleets and bounded costs

The exact two-service example isolates the mechanism; it does not measure its prevalence or size in service. For a timetable-scale check we use the published Hildenbrand instance from the Sistig electric-bus scheduling data [Sistig et al., 2025a,b]: 37 mandatory services with a complete directed deadhead matrix. We form two *separate* one-depot cases, using depot 15 or depot 16. These are declared modifications of the source network, not a reconstruction of its two-depot, multiple-output charging design. Timetable and directed travel times retain their source offsets, including the next-day services. Matrix travel times depend on origin and destination, not on departure time.

Energy and monetary inputs are explicit scenario assumptions, not observed vehicle telemetry or source objective coefficients. Service energy is 1.20 kWh/km plus 16 kWh per service hour. Directed deadheads use 0.96 kWh/km plus 16 kWh per travel hour, and explicit off-depot waiting uses 16 kWh/h; depot-parked auxiliary draw is zero. Each used bus has 400 kWh usable inventory, zero additional reserve, unit charging efficiency, and a 360 kW acceptance limit. One 360 kW connector is shared at the selected depot. All used buses start full and return full by a synthetic 30:00 service-block boundary. Charging has no taper and no switching time. This finite block does not establish a repeating daily operation. A 37-bus one-service-per-bus construction confirms feasibility of both cases, but has no optimization role. Monetary travel cost is zero: the objective charges the used-bus fee and grid energy, while movement energy and time remain in the physical constraints.

We first use synthetic used-bus cost $f = 100$ and flat grid price $a = 0.2$ per kWh. Because this objective is linear in load, minimizing it over the hull of complete plans has the same value as minimizing it over physical plans; this flat-price experiment cannot exhibit a physical-hull gap. The original fixed-budget native runs returned physically replayed two-bus schedules in both depot cases without proving objective optimality. A separate, already archived round-0 solve within the later nonlinear planner used the same complete depot-15 physical case and a tangent objective that reduces exactly to the flat price. That call returned a *native numerical* optimum of approximately 408.53, with matching incumbent and lower bound under solver tolerances. This is a retrospective interpretation of one call, not a new flat-price experiment or a reclassification of the nonlinear

campaign. The numerical certificate does not identify the exact rational-model optimum. The following ideal stored-input certificates answer that distinct exact-arithmetic question.

For both depot graphs, a one-bus path is impossible. Chronologically forced adjacent services leave no admissible depot visit, while mandatory service energy alone exceeds one bus’s 400 kWh inventory. An exact cardinality-gated path-cover relaxation of each full 37-service graph gives flat-objective lower bounds displayed outward as 404.92 at depot 15 and 414.39 at depot 16. The bounds retain every declared connection mode but relax some physical constraints. Separately, an exact rational depot-15 witness covers all 37 services with two buses, respects the shared connector, and restores both batteries to 400 kWh by 30:00. Hence depot 15’s ideal model has a minimum fleet of two and the *flat-cost optimum* lies in

$$D_{15}^{\text{flat}} \in [404.92, 408.54] \quad (\text{outward-rounded exact rational bounds}). \quad (13)$$

The bounds do not identify the optimum. Depot 16 has the exact at-least-two obstruction and lower bound, but its returned two-bus upper witness remains numerical, yielding a mixed-evidence interval $[414.39, 433.75]$ after outward rounding. In particular, a native solver’s rounded matrix and its feasibility tolerances are distinct from the ideal stored-input rational model.

For a nonlinear illustration on *the same finite depot-15 ideal model*, let e_t be hourly grid energy for 30 market intervals and set

$$F(e) = \sum_{t=1}^{30} \left(0.2e_t + \frac{e_t^2}{1800} \right). \quad (14)$$

The coefficients and bus cost remain synthetic. The exact two-bus witness gives a physical upper bound displayed as 512.77. To bound the complete hull from below, let L_{flow} denote the exact depot-15 two-path flat-cost relaxation value. Because monetary travel cost is zero, it implies at least $E_{\text{min}} = (L_{\text{flow}} - 2f)/a \approx 1024.62$ kWh for any two-bus plan. One bus is impossible, and a uniform-price comparison using the service-energy floor excludes three-or-more-bus plans from improving the dual lower certificate. The 30-period Fenchel inequality then gives the exact rational lower bound displayed as 424.36. Therefore

$$424.36 \leq CH_{15} \leq D_{15} \leq 512.77, \quad 0 \leq D_{15} - CH_{15} \leq 88.41. \quad (15)$$

All displayed endpoints are rounded outward from the exact bounds. They do not establish CH_{15} or D_{15} individually, nor a strictly positive gap. The public case thus demonstrates physically feasible complete-fleet witnesses and useful bounds under declared assumptions, while leaving public price support open.

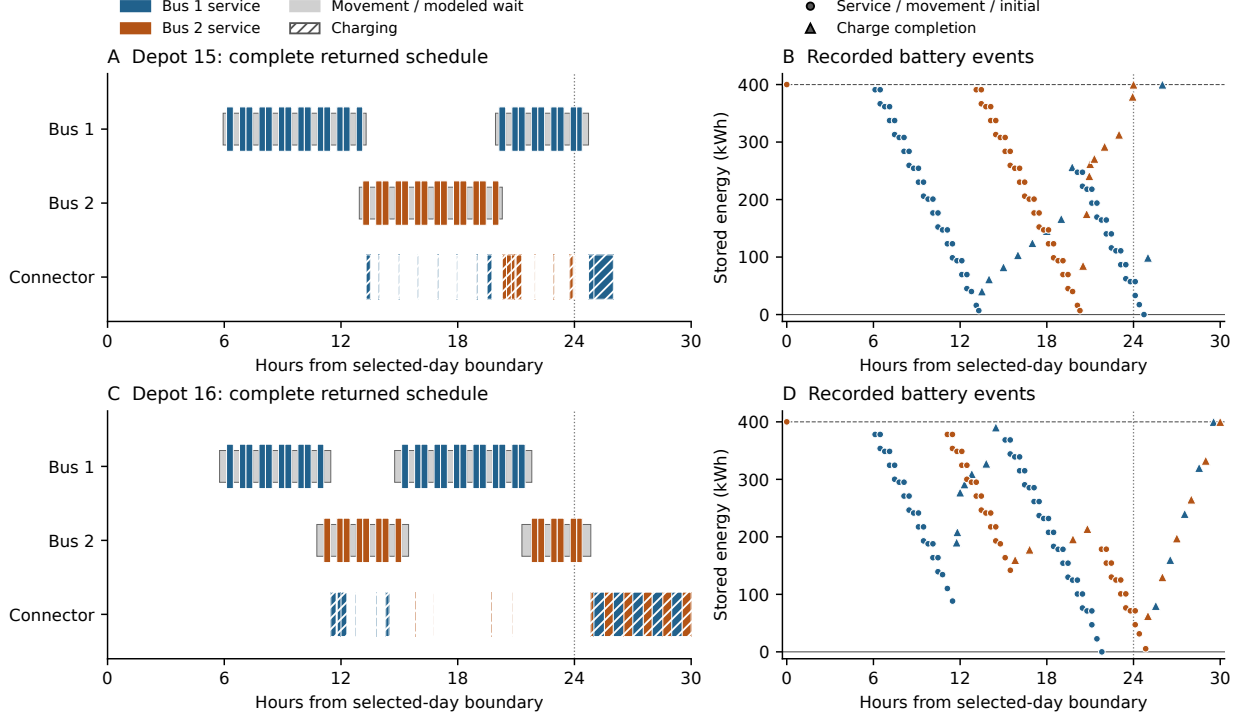


Figure 3: Two-bus schedules returned by the original flat-price pilot for the two depot variants of the same Hildenbrand timetable. Battery and charging traces describe numerical feasible witnesses under the declared scenario assumptions. They are not measured operation, two independent networks, or proofs of cost optimality. The separately constructed exact depot-15 witness supplies the ideal-model bounds in the text.

6 Computational price-support diagnostics

The computational question is whether a physical optimum is supported by its own marginal price. A useful diagnostic therefore reports the physical value D , hull value CH , their nonnegative difference and own-price regret together. We first strengthen the public lower bounds analytically, then summarize an existing development screen. All cases in that screen are development cases; two depot variants still represent one public timetable.

6.1 An analytical baseline for both markets

Use the public physical cases in Section 5. Market 0 has $a_{0,t} = 0.20$ throughout; market 1 has $a_{1,t} = 0.18$ for the first 15 hours and 0.22 for the last 15. Both have $b = 1/900$ in $F_s(e) = \sum_t (a_{s,t}e_t + be_t^2/2)$. This changes only supply cost, leaving fleet feasibility fixed. As in the existing exact public certificates, arithmetic uses the stored input coefficients as rationals; the decimal parameters here describe those inputs to their displayed precision.

Let $E_{\min}$ be the two-bus total-energy floor obtained from the exact flow relaxation, and let E_{svc} be mandatory service energy. With full replenishment, efficiency one and zero monetary travel cost, $E_{\min} = (L_{\text{flow}} - 2f)/a_0$. This gives about 1024.62 and 1071.97 kWh for depots 15 and 16, respectively; E_{svc} is about 889.12 kWh in both. The one-bus obstruction is unchanged. For a

Table 1: Analytical hull lower bounds for the ideal public model, rounded down to two decimals. The last column is a numerical lower bound from the separate later solver-baseline comparison, included only to expose the missing analytical baseline. No runtime comparison or exact native-model certificate is implied.

Depot	Analytical, market 0	Analytical, market 1	Later computed, market 1
15	424.36	418.96	405.83
16	435.67	430.27	420.45

uniform price $p > 0$, every two-bus plan has $c + p \sum_t e_t \geq 2f + pE_{\min}$, whereas every plan using at least three buses has $c + p \sum_t e_t \geq 3f + pE_{\text{svc}}$. Hence, whenever

$$3f + pE_{\text{svc}} \geq 2f + pE_{\min}, \quad (16)$$

the lower bound $2f + pE_{\min}$ holds for every physical plan and its convex hull. For $p \geq \max_t a_{s,t}$, Fenchel’s inequality then gives

$$CH_s \geq 2f + pE_{\min} - \sum_{t=1}^{30} \frac{(p - a_{s,t})^2}{2b}. \quad (17)$$

The maximizing uniform test price is $p_s^* = \bar{a}_s + bE_{\min}/30$, where $\bar{a}_s = \sum_t a_{s,t}/30$. For both markets it exceeds 0.22 in each depot case, and the left minus right side of (16) is strictly positive: about 67.76 at depot 15 and 56.17 at depot 16. Thus no unexamined fleet cardinality invalidates the bound.

Table 1 recovers the existing depot-15 original-market certificate and extends it to the other depot and market. In the changed market, the exact floors exceed the later numerical lower endpoints by about 13.13 and 9.82 cost units. These inexpensive bounds should be included before assessing what iterative optimization adds. The hypothetical higher floor obtained by substituting a native flat optimum would retain native model and tolerance qualifications; it is not used as an exact certificate.

6.2 Physical cost, hull cost and incumbent regret

The initial screen used the same two public cases and two small synthetic cases: the two-service cyclic construction and a three-service case with multiple depot visits. Each was evaluated in an original and a changed market. Table 2 shows all four changed-market cases, using the screen’s physical planner, cold hull calculation and response at the returned planner incumbent’s own price, combined after the run. Public planner and hull stages had nominal 180-second wall allowances and 160-second native phases; responses had 60 seconds. Synthetic planner and hull stages had 60-second wall allowances with 45-second native phases, and responses had 30 seconds. Complete wrapper allowances were recorded separately. Call and arithmetic limits also restricted the calculations. In the four-hour synthetic target markets, the cyclic case uses $a_1 = (0, 3.8, 0, 0.2)$ and $b = (0, 0.2, 0, 0.2)$; the multivisit case uses $a_1 = (0.18, 0.18, 0.22, 0.22)$ and $b = (0.2, 0.2, 0.2, 0.2)$. The archive fixes their physical inputs.

For the specific planner incumbent $\hat{x}$, let $B_{\hat{x}} = c(\hat{x}) + p_{\hat{x}}^\top e(\hat{x})$. A response enclosure $V(p_{\hat{x}}) \in [L_V, U_V]$ gives $r(\hat{x}) \in [B_{\hat{x}} - U_V, B_{\hat{x}} - L_V]$. The stored response price matches the gradient computed for that incumbent. We also use the response lower bound to improve the hull lower bound through $L_V - F_s^*(p_{\hat{x}})$, and any physical upper witness as a hull upper witness. The physical lower bound

Table 2: Changed-market support diagnostics from the initial development screen, with same-screen native bounds combined after the run. All endpoints are rounded outward. Regret belongs to the returned planner incumbent $\hat{x}$ in that row; public planner optima are unknown. Analytical ideal-model energy floors are reported separately in Table 1.

Case	Physical D	Hull CH	Gap $D - CH$	Regret $r(\hat{x})$
Two-service cyclic	[98.99, 99.01]	[97.98, 97.99]	[1.01, 1.02]	[8.99, 9.01]
Three-service visits	[84.71, 84.73]	[84.62, 84.73]	[0.00, 0.10]	[0.09, 0.10]
Public depot 15	[339.35, 517.33]	[339.35, 517.33]	[0.00, 177.98]	[190.43, 235.09]
Public depot 16	[377.65, 547.88]	[377.65, 547.88]	[0.00, 170.22]	[215.95, 269.16]

cannot be weaker than the hull lower bound. The table combines these same-screen inequalities before computing the gap interval; it claims no reduction in the original algorithm’s time. These are native numerical results: they retain solver and physical-replay tolerances and use rounded prices. Exact processing of their stored values does not make them ideal-model proofs.

The public rows establish sizeable incentives to change the *returned* schedules under their fixed own prices. They do not establish regret of an optimal public fleet: the planner bounds remain open, so these schedules may simply be costly incumbents. The gap intervals include zero and do not certify that the public timetable lacks a supporting physical plan.

The multivisit response supplies a hull lower bound of 84.62, reducing the same-screen planning-gap upper bound to 0.10 without another optimization. This is a consequence of combining already available certificates, not a measured benefit of reuse.

Postprocessing the same public planner uppers with the exact energy floors from Table 1 reduces the changed-market gap caps to 98.37 at depot 15 and 117.61 at depot 16. These are mixed-evidence bounds, conditional on the native upper witnesses representing feasible plans of the ideal model; the lower floors themselves are exact. They are not timed results of the original algorithm. The original-market exact depot-15 certificate in (15) remains separate and stronger for that market.

6.3 What the computations establish

Table 3: Evidence supporting the current conclusions.

Finding	Scope and remaining question
Exact support obstruction	A fully replenished synthetic fleet and a specified robust variant have a positive gap; replication separates the planning gap from whole-operator regret.
Analytical public bounds	Flow, cardinality and energy inequalities give exact ideal-model lower bounds in two markets. Both depot variants share one timetable.
Incumbent public incentives	Native responses improve the bill of each named screen incumbent at its own price. Regret at a planner optimum and a positive public planning gap remain unresolved.
Repeated-solve acceleration	Development comparisons are preserved in the supplement. Cap dependence, software failures and unconfirmed start acceptance prevent a general speedup claim. Retrieval and learned proposals remain untested.

The computation presently demonstrates useful bounds and incumbent incentives, with the limits in Table 3. It does not measure the prevalence of price-support failure across transport networks. Further work must improve physical-pricing bounds, compare methods at common quality and scale to independent timetables before making general computational claims.

7 Implications and limits

The analytical and computational parts answer complementary questions. The construction proves that continuous charging and full replenishment do not remove the possibility of a marginal-price support failure. The experiments measure how difficult it is to obtain informative full-fleet bounds on declared instances. An unresolved public interval neither confirms the analytical mechanism on that timetable nor refutes it. Likewise, a strong restricted-master solution says little about the cost of discovering an ungenerated physical fleet unless global pricing has supplied a corresponding bound.

The analytical energy floor is a baseline that any computational method should use when bounding this model. A weak solver lower bound should not displace it. Strengthening physical pricing remains useful for narrowing global intervals, but the underlying cardinality and energy inequalities must be translated into each pricing model with their assumptions intact. A lower bound on the nonlinear hull objective is not by itself a valid cut on every linear price-response objective.

The exploratory comparisons in the development supplement do not establish general acceleration. On a three-service case, cold runs stopped at call or arithmetic limits well before their wall-time allowance; success from a retained pool is therefore conditional on those caps. Public solver-start comparisons used one seed and did not establish that submitted starts were accepted. Future experiments should size algorithmic limits before fixing the evaluation design, log start acceptance, compare time to common solution quality across repeated seeds, and include independent timetables. Retained complete fleets and nearest-neighbor retrieval should precede learned proposals, with direct proposal quality, physical repair and global verification reported separately. All variants of a base timetable must stay in one data group, with independent test timetables reserved before training.

Failure of a particular price-update iteration would not itself prove that no supporting price exists. Such convergence experiments must retain the same physical model, including terminal replenishment, and must be distinguished from the support and regret statements proved here.

The physical model abstracts from stochastic delays, battery degradation, temperature-dependent consumption, nonlinear charging taper and operational switching times. Public service and travel inputs are combined with declared energy and economic assumptions; the results are not estimates of savings for the source operator. Resource rights also delimit the economic interpretation. A whole-fleet price-taking deviation, an individually reserved-pair deviation and a strategic operator anticipating its own price impact are different optimization problems. Conclusions must follow the chosen participant and settlement model.

8 Conclusion

A smooth convex electricity supply cost need not support an indivisible electric-bus fleet at its own marginal prices, even with continuous charging and full terminal replenishment. Replication

can make the planning gap small without making the incentive of a whole-fleet operator small along every subsequence, while incentives for reserved pairs vanish. Analytical cardinality and energy bounds strengthen the public computations. Joint physical, hull and regret reporting leaves public price support unresolved and separates incumbent incentives from optimal-fleet incentives. Establishing the mechanism’s practical prevalence and an effective computational method requires further independent timetable evidence.

Data and reproducibility

The EGG research repository contains the analytical constructions, declared public-data transformation, experiment protocols, complete outcome tables including unsuccessful attempts, and figure-generation sources. A companion source map links the manuscript’s claims to reviewed evidence and distinguishes exact ideal-model calculations from tolerance-qualified native results. All cases reported here are development cases. No private operational data or reserved evaluation outcomes are used. The figures in this manuscript are drawn from already reviewed results; its assembly required no new optimization. The separate development supplement retains the earlier computational comparisons, including failures and their resource use, without treating them as a validated acceleration study.

A Proofs for the complete-fleet statements

Proof of Proposition 1. For physical x and any hull point (e, c) , linear minimization over the convex hull gives $c + p_x^\top e \geq V(p_x)$. Convexity gives $F(e) \geq F(e(x)) + p_x^\top (e - e(x))$. Adding and minimizing over the hull yields $CH \geq c(x) + F(e(x)) - r(x)$, proving the first inequality; $c(x) + F(e(x)) \geq D \geq CH$ proves the second. If $r(x) = 0$, all inequalities collapse to $CH = D = c(x) + F(e(x))$. Conversely, if $CH = D$, compactness supplies a physical minimizer x^* . It is also a hull minimizer. For every physical y , the directional derivative of $c + F(e)$ at $(e(x^*), c(x^*))$ toward $(e(y), c(y))$ is nonnegative: $c(y) + p_{x^*}^\top e(y) \geq c(x^*) + p_{x^*}^\top e(x^*)$. Thus $r(x^*) = 0$. The same argument applies to every physical minimizer. $\square$

For the accounting identity, $F^*(p) \geq p^\top e(x) - F(e(x))$ by definition, so supplier LOC is non-negative. Substitution in (4) gives the claimed sum. A compact hull and a supply cost differentiable near it yield a supporting price p^* at a hull minimizer (e^*, c^*) ; linear first-order optimality gives $V(p^*) = c^* + (p^*)^\top e^*$, and Fenchel equality gives $F^*(p^*) = (p^*)^\top e^* - F(e^*)$. Hence $V(p^*) - F^*(p^*) = CH$. At a physical planner optimizer the LOC sum is $D - CH$. At the plan’s own gradient price, Fenchel equality makes supplier LOC zero.

Proof of (6). Let (e^*, c^*) be a hull minimizer, $p^* = a + Qe^*$, $d = \|e(x) - e^*\|_Q$, and $H = H(x)$. First-order hull optimality makes (e^*, c^*) a minimizer of $c + (p^*)^\top e$, so

$$\text{LOC}_{\text{fleet}}(x; p^*) = H - \frac{1}{2}d^2 \geq 0.$$

In particular $d \leq \sqrt{2H}$. For each physical alternative y , the change in its linear advantage when the price moves from p^* to $p_x = p^* + Q(e(x) - e^*)$ is $(e(x) - e^*)^\top Q(e(x) - e(y))$, at most dR_Q by the seminorm Cauchy–Schwarz inequality. Maximizing over y gives $r(x) \leq H - d^2/2 + dR_Q \leq H + R_Q\sqrt{2H}$. The lower inequality is Proposition 1. $\square$

Proof of Proposition 2. Let λ be the one-bus mixture weight and x the mean early load. Complete-plan mixtures have $0 \leq \lambda \leq 1$, $10\lambda \leq x \leq 10$, and intrinsic cost $14 - 7\lambda$. These conditions are sufficient: for $\lambda < 1$ the two-bus component buys $(x - 10\lambda)/(1 - \lambda) \in [0, 10]$ early; at $\lambda = 1$, $x = 10$. For fixed x , cheapest intrinsic cost uses $\lambda = x/10$. Substituting $z = 30 - x$ into (7) gives hull objective

$$14 - \frac{7x}{10} + F(x, 30 - x) = 104 - 2.7x + 0.2x^2 = \frac{7591}{80} + \frac{(x - 27/4)^2}{5}.$$

Its minimizer is $x = 27/4$, $\lambda = 27/40$. Direct minimization of the one- and two-bus physical branches gives respectively 97 and 99. At $(10, 20)$ the gradient of F is $(6, 4)$; its private objective is $7 + 6(10) + 4(20) = 147$, while the two-bus $x = 0$ plan has $14 + 4(30) = 134$. At hull load $(27/4, 93/4)$ the gradient is $(107/20, 93/20) = (5.35, 4.65)$. Both supporting plans have private objective 153.5, and $F^*(p) = 58.6125$ on $\mathbb{R}_+^2$. The hull dual value is therefore 94.8875. The physical one-bus plan is a price minimizer, so fleet LOC is zero; supplier LOC is $97 - 94.8875$. $\square$

Proof of Proposition 3. For reserve $0 \leq r \leq 5$, efficiency η , one-hour early limit P , and one-hour single terminal connector K , write

$$T = 30/\eta, \quad \ell = \max\{0, T - K\}, \quad u = \min\{P, 15/\eta\}, \quad h = \max\{\ell, (10 + r)/\eta\}.$$

The two-bus physical early-energy interval is $[\ell, u]$, and the one-bus interval is $[h, u]$ when nonempty. The lower limit ℓ enforces terminal connector capacity, while $(10 + r)/\eta$ ensures the one bus can begin B above reserve. For every point in these intervals, charge only A's bus early; terminal demands total $T - x \leq K$ and can be served serially in the one-hour terminal window. Individual inventories and acceptance limits hold, and all used buses finish at capacity.

Write $G_T(x) = F(x, T - x)$, $k = 1/5$, and $m = T/2 - 5a/2$ for early linear coefficient $a = 4$. Then $G_T(x) = G_T(m) + k(x - m)^2$. Suppose $\ell < m < h < u$, $f > 0$, and $f < 2k(h - \ell)(h - m)$. The cheapest hull operating-cost boundary over $[\ell, h]$ is $2f - f(x - \ell)/(h - \ell)$, obtained by mixing the two-bus endpoint ℓ and one-bus endpoint h ; over $[h, u]$ it is f . Put $d = h - \ell$. The strict inequality places the hull minimum $x^* = m + f/(2kd)$ inside (m, h) . Physical branch minimization gives

$$D = G_T(m) + \min\{f + k(h - m)^2, 2f\}, \quad CH = G_T(m) + 2f - \frac{f(m - \ell)}{d} - \frac{f^2}{4kd^2}.$$

Consequently

$$\Delta = \min \left\{ k(h - x^*)^2, \frac{f(m - \ell)}{d} + \frac{f^2}{4kd^2} \right\} > 0.$$

At $(r, \eta, P, K, f, a) = (1, 19/20, 12, 30, 7, 4)$, $(\ell, m, h, u, d) = (30/19, 110/19, 220/19, 12, 10)$. The inequalities are strict: $u - h = 8/19$ and $2kd(h - m) = 440/19 > 7$. Rational substitution yields (9). All strict inequalities persist under small changes in (r, η, P) with $K = 30$ fixed. $\square$

Proof of Proposition 4. For m paired buses, total early energy E ranges over $[10m, 10n]$, with intrinsic cost $14n - 7m$ and terminal energy $30n - E$. Convexifying complete plans gives a triangular projected set; its cheapest operating cost at each E is $14n - 0.7E$. Substitution in (10) gives a quadratic minimized at $E = 27n/4$ with value $7591n/80$.

For fixed integer $m \geq n/2$, physical supply cost is minimized at $E = 10m$, and the total objective is $104n - 27m + 20m^2/n = 7591n/80 + 20(m - 27n/40)^2/n$. For $m \leq n/2$, the best early load is $E = 5n$ when feasible and the branch objective is $99n - 7m$, decreasing toward $n/2$. For odd n , moving from $m = \lfloor n/2 \rfloor$ to $m = \lceil n/2 \rceil$ changes the minimized objective by $-7 + 5/n < 0$.

For even n , the two formulas agree at $m = n/2$, and an integer nearest to $27n/40$ lies in the upper regime. Thus the nearest integer to $27n/40$ is optimal, with both integers at a half-integer tie and $|\delta| \leq 1/2$.

At a physical optimum, the price is $p = (4 + 2m/n, 6 - 2m/n)$. The one-bus private objective minus the best two-bus private objective is $40\delta/n$; the latter buys zero early energy when $m/n \geq 1/2$. Hence whole-operator regret is

$$r_n = \begin{cases} m 40\delta/n, & \delta \geq 0, \\ (n - m)(-40\delta/n), & \delta < 0. \end{cases}$$

For $n = 40k + 1$, $m = 27k + 1$ and $\delta = 13/40$, giving (12). The same difference is a reserved pair's individual regret when its current choice is worse; $|40\delta/n| \leq 20/n$. Summing the disadvantaged pairs reproduces the whole-operator expression because each pair has its own reserved connector rights. $\square$

References

- Panagiotis Andrianesis, Dimitris Bertsimas, Michael C. Caramanis, and William W. Hogan. Computation of convex hull prices in electricity markets with non-convexities using Dantzig–Wolfe decomposition. *IEEE Transactions on Power Systems*, 37(4):2578–2589, 2022. doi: 10.1109/TPWRS.2021.3122000. URL <https://doi.org/10.1109/TPWRS.2021.3122000>. Author manuscript, arXiv:2012.13331v2.
- George B. Dantzig and D. R. Fulkerson. Minimizing the number of tankers to meet a fixed schedule. *Naval Research Logistics Quarterly*, 1(3):217–222, 1954. doi: 10.1002/nav.3800010309. URL <https://onlinelibrary.wiley.com/doi/10.1002/nav.3800010309>.
- Marelot H. de Vos, Rolf N. van Lieshout, and Twan Dollevoet. Electric vehicle scheduling in public transit with capacitated charging stations. *Transportation Science*, 58(2):279–294, 2024. doi: 10.1287/trsc.2022.0253. URL <https://doi.org/10.1287/trsc.2022.0253>. Published online 12 December 2023.
- Lingwen Gan, Ufuk Topcu, and Steven H. Low. Optimal decentralized protocol for electric vehicle charging. *IEEE Transactions on Power Systems*, 28(2):940–951, 2013. doi: 10.1109/TPWRS.2012.2210288. URL <https://doi.org/10.1109/TPWRS.2012.2210288>.
- Paul R. Gribik, William W. Hogan, and Susan L. Pope. Market-clearing electricity prices and energy uplift, December 2007. URL https://www.lmpmarketdesign.com/papers/Gribik_Hogan_Pope_Price_Uplift_123107.pdf. Working paper, 31 December 2007.
- Bowen Hua and Ross Baldick. A convex primal formulation for convex hull pricing. *IEEE Transactions on Power Systems*, 32(5):3814–3823, 2017. doi: 10.1109/TPWRS.2016.2637718. URL <https://doi.org/10.1109/TPWRS.2016.2637718>.
- Lukas Hümbes, Alexander Martin, and Lars Schewe. Exploiting complete linear descriptions for decentralized power market problems with integralities. *Mathematical Methods of Operations Research*, 95:451–474, 2022. doi: 10.1007/s00186-022-00775-z. URL <https://doi.org/10.1007/s00186-022-00775-z>.

- Mehdi Madani, Carlos Ruiz, Sauleh Siddiqui, and Mathieu Van Vyve. Convex hull, IP and european electricity pricing in a european power exchanges setting with efficient computation of convex hull prices, 2018. URL <https://arxiv.org/abs/1804.00048v1>. Preprint, version 1.
- Richard P. O’Neill, Paul M. Sotkiewicz, Benjamin F. Hobbs, Michael H. Rothkopf, and William R. Stewart. Efficient market-clearing prices in markets with nonconvexities. *European Journal of Operational Research*, 164(1):269–285, 2005. doi: 10.1016/j.ejor.2003.12.011. URL <https://doi.org/10.1016/j.ejor.2003.12.011>.
- Axel Parmentier, Rafael Martinelli, and Thibaut Vidal. Electric vehicle fleets: Scalable route and recharge scheduling through column generation. *Transportation Science*, 57(3):631–646, 2023. doi: 10.1287/trsc.2023.1199. URL <https://doi.org/10.1287/trsc.2023.1199>.
- Dane A. Schiro, Tongxin Zheng, Feng Zhao, and Eugene Litvinov. Convex hull pricing in electricity markets: Formulation, analysis, and implementation challenges. *IEEE Transactions on Power Systems*, 31(5):4068–4075, 2016. doi: 10.1109/TPWRS.2015.2486380. URL <https://doi.org/10.1109/TPWRS.2015.2486380>.
- Hubert Maximilian Sistig, Philipp Sinhuber, Matthias Rogge, and Dirk Uwe Sauer. Dataset for evaluating costs and operations of public bus fleet electrification, 2025a. URL <https://doi.org/10.6084/m9.figshare.26088190.v1>. Version 1; CC BY 4.0.
- Hubert Maximilian Sistig, Philipp Sinhuber, Matthias Rogge, and Dirk Uwe Sauer. Evaluating costs and operations of public bus fleet electrification. *npj Sustainable Mobility and Transport*, 2:15, 2025b. doi: 10.1038/s44333-025-00030-y. URL <https://doi.org/10.1038/s44333-025-00030-y>.
- Ross M. Starr. Quasi-equilibria in markets with non-convex preferences. *Econometrica*, 37(1):25–38, 1969. doi: 10.2307/1909201. URL <https://doi.org/10.2307/1909201>.
- Weitiao Wu, Yue Lin, Ronghui Liu, and Wenzhou Jin. The multi-depot electric vehicle scheduling problem with power grid characteristics. *Transportation Research Part B: Methodological*, 155: 322–347, 2022. doi: 10.1016/j.trb.2021.11.007. URL <https://doi.org/10.1016/j.trb.2021.11.007>.